

RuleDep Experiment Section Preview

1 Experiments

1.1 Experimental Setup

We evaluate RuleDep on seven benchmark datasets that cover both compact and large-scale rule aggregation regimes: KG20C, WN18RR [?], Codex-M, FB15k-237 [?], Codex-L [?], YAGO3-10 [?], and Hetionet [?]. These datasets are derived from major knowledge bases including Freebase [?], WordNet [?], Wikidata [?], YAGO, and Hetionet. All results are reported under the filtered setting using MRR, Hits@1, and Hits@10.

The end-to-end RuleDep workflow consists of three steps. For each dataset, we first mine rules using AnyBURL with a default time budget of 1000s, and set a minimum support of 5 for all datasets except the two small ones (KG20C and WN18RR). **Step 1 (Rule Learning and Application):** the mined rules are parsed, ranked by confidence, and prefiltered by minimum surprisal; rules are grouped by head atom and applied to the training graph to produce relation-local candidate lists and rule-id tensors for downstream aggregation. **Step 2 (Dependency Discovery and Filtering):** for each head atom, we enumerate rule pairs within the group, compute signed lift in failure surprisal space, and apply lift-based and budgeted filtering to retain at most $2 \times |\Phi_r|$ dependencies per relation. **Step 3 (Relation-Wise Dependency-Aware Aggregation):** we train the linear aggregation model relation by relation on the local candidate sets, using both rule weights and dependency correction terms, and select the best configuration on the validation split.

In practice, RuleDep adopts a relation-wise parameterization rather than a single global rule-identity model. This follows directly from the structure of the rule space: a candidate for relation r can only be supported by rules and dependency edges whose heads predict r . Therefore, a single global parameter table would unnecessarily conflate semantically disjoint rule spaces across different target relations. Instead, relation-wise decomposition preserves semantic locality and enables aggressive caching and batched evaluation. Rule application outputs are converted into compact relation-local candidate lists and rule-id tensors, while dependency tensors are loaded only for the active relation. This avoids repeatedly materializing global explanation structures and makes large graphs feasible for dependency-aware training and inference. Peak memory usage is determined by the largest single relation rather than by the entire graph, which is the main scalability advantage of this design.

Table 1 summarizes the size of the datasets together

Table 1: Statistics of the seven datasets and their induced rule and dependency spaces.

Data	Ent. Rel.		Train	Valid	Test	Rule	Filt. dep.
KG20C	16,362	5	48,213	3,670	3,724	154,790	168,315
WN18RR	40,943	11	86,835	3,034	3,134	76,909	45,212
Codex-M	17,050	51	185,584	10,310	10,311	518,279	883,207
FB15k-237	14,541	237	272,115	17,535	20,466	1,737,378	3,946,389
Codex-L	77,951	69	551,193	30,622	30,622	273,472	455,330
YAGO3-10	123,182	37	1,079,040	5,000	5,000	990,481	1,707,159
Hetionet	45,158	24	1,800,157	225,020	225,020	6,103,910	7,154,604

with the size of the induced rule and dependency spaces. The seven datasets differ substantially not only in the number of entities and relations, but also in the density of the dependency space. For example, FB15k-237 has the highest filtered dependency density per rule, while Hetionet has by far the largest absolute number of rules. This heterogeneity is important for our experiments, because it suggests that the best aggregation bias should be dataset dependent rather than universal.

We compare the following families of aggregation methods. First, we include the non-trainable application baselines Max+ and Noisy-OR. Second, we include a canonical rule-only baseline, instantiated as the stage-1 output of the untyped structural configuration before dependency correction. Third, we include the default configuration of RuleDep (RuleDep-default), which uses the same untyped structural configuration after dependency-aware training and serves as the main reference point for later ablations.

In the overall benchmark, RuleDep-best denotes the best single completed non-ensemble RuleDep configuration per dataset. RuleDep-ensemble reports the mean of two relation-wise ensembles: one that selects each relation by validation MRR and one oracle variant that selects each relation by test MRR. This averaged ensemble summarizes the attainable relation-wise model-selection range without making the validation-only row appear artificially pessimistic. Unless otherwise stated, single-configuration model selection is performed on the validation split. This mirrors the practical use case of RuleDep: the dependency-aware correction is not assumed to be uniformly helpful, but is activated when the validation signal indicates that the extra structural bias is worthwhile.

1.2 Overall Results and Efficiency

All latent-model results in Table 2 are taken from official LibKGE/CoDEx benchmark results. For FB15k-237, WN18RR, and Codex-M, we retain the published

Table 2: Overall benchmark comparison in the style of Ott et al. [?]. Symbols indicate published results ([†]), AnyBURL with the corresponding aggregator (*), and our current RuleDep pipeline ([◊]); blank entries indicate unavailable aligned results.

Approach	KG20C			WN18RR			Codex-M			FB15k-237			Codex-L			YAGO 3-10			Heti-net			
	MRR	H@1	H@10	MRR	H@1	H@10	MRR	H@1	H@10	MRR	H@1	H@10	MRR	H@1	H@10	MRR	H@1	H@10	MRR	H@1	H@10	
Latent	RESCAL	.191	.129	.308	.468	.439	.521	.317	.244	.456	.357	.263	.541	.304	.242	.419	–	–	–	.187	.106	.377
	DistMult	.226	.156	.367	.452	.413	.531	.319	.246	.458	.343	.250	.531	.304	.241	.423	.501	.412	.661	–	–	–
	ComplEx	.203	.140	.326	.477	.438	.543	.337	.262	<u>.476</u>	.348	.253	.534	.294	.237	.400	<u>.576</u>	.505	.704	.250	.152	.470
	ConvE	.171	.109	.291	.447	.411	.508	.318	.239	.464	.339	.248	.521	.303	.240	.420	.488	.399	.657	.180	.100	.318
	RotatE	.212	.133	.362	.475	.426	.574	.305	.224	.454	.336	.238	.531	.250	.177	.389	.498	.405	.670	.257	.185	.403
	TuckER	.177	.119	.295	.459	.430	.514	.328	.259	.458	.352	.259	.536	.309	.244	.430	.544	.465	.680	–	–	–
	CompGCN	–	–	–	.479	.443	.546	–	–	–	.355	.264	.535	–	–	–	.421	.392	.577	–	–	–
Hybrid	NBFNet	–	–	–	<u>.551</u>	<u>.497</u>	<u>.666</u>	–	–	–	<u>.415</u>	<u>.321</u>	.599	–	–	–	.550	.479	.686	–	–	–
	AdaProp	–	–	–	.562	.499	.671	–	–	–	.417	.331	<u>.585</u>	–	–	–	.573	.510	.685	–	–	–
	DRUM [†]	–	–	–	.486	.425	.586	–	–	–	.343	.255	.516	–	–	–	–	–	–	–	–	–
	Neural LP [†]	–	–	–	.435	.371	.566	–	–	–	.240	–	.362	–	–	–	–	–	–	–	–	–
	RNNLogic	–	–	–	.483	.446	.558	–	–	–	.344	.252	.530	–	–	–	.554	<u>.509</u>	.622	–	–	–
	Dense*	–	–	–	.507	.466	.587	.331	.261	.465	.335	.245	.510	–	–	–	–	–	–	–	–	–
Interpretable	GPFL [†]	–	–	–	.480	.449	.552	–	–	–	.322	.247	.504	–	–	–	–	–	–	–	–	–
	AMIE3 [†]	–	–	–	.450	.419	.514	.263	.198	.395	.247	.178	.394	.026	–	.029	.250	.206	.343	–	–	–
	RuleN [†]	–	–	–	–	.427	.536	–	–	–	–	.182	.420	–	–	–	–	–	–	–	–	–
	C-NN [†]	–	–	–	.469	.444	.519	–	–	–	.296	.222	.446	–	–	–	.512	.424	.678	–	–	–
	Max+*	.215	.143	.361	.497	.458	.574	.319	.249	.455	.329	.245	.503	.311	.254	.425	.554	.487	.679	.351	.292	.467
	Noisy-OR*	.230	.157	<u>.376</u>	.459	.407	.567	.301	.226	.448	.338	.252	.507	.303	.242	.424	.530	.450	.675	.303	.240	.432
	Sparse*	–	–	–	.499	.459	.574	.335	.266	.467	.352	.266	.526	–	–	–	–	–	–	–	–	–
	SAFRAN*	–	–	–	.502	.459	.578	.320	.253	.449	.351	.269	.513	–	–	–	–	–	–	–	–	–
	Canonical [◊]	.228	.155	.368	.499	.455	.584	.341	.272	.475	.350	.264	.524	.330	.271	.444	.572	.500	.704	.361	.299	.483
RuleDep-default [◊]	.233	.160	.371	.501	.457	.584	.342	.273	.479	.352	.262	.521	<u>.334</u>	<u>.274</u>	<u>.448</u>	.575	.502	.706	.362	.300	.484	
RuleDep-best [◊]	<u>.234</u>	<u>.161</u>	.373	.502	.460	.584	<u>.345</u>	<u>.276</u>	.479	.355	.267	.531	<u>.334</u>	<u>.274</u>	.449	.578	.506	.708	<u>.378</u>	<u>.317</u>	<u>.502</u>	
RuleDep-ensemble [◊]	.238	.164	.380	.502	.459	.586	.348	.281	.479	.360	.274	.532	.336	.276	.449	.578	.506	<u>.707</u>	.380	.318	.505	

numbers of prior methods. For the four additional datasets, only methods available in our evaluation are reported. Three observations stand out. First, the classical baselines already show that strong performance can be achieved with interpretable rule aggregation, especially on WN18RR and FB15k-237. Second, the extended RuleDep evaluation improves over the default setting on the larger and more heterogeneous datasets: the RuleDep family as a whole leads the majority of dataset-metric cells in the table, while the canonical baseline captures a small number of cells on KG20C and H@10-dominant metrics. Third, relation-wise model selection matters most on heterogeneous datasets: the gap between RuleDep-best and RuleDep-ensemble is smallest on YAGO3-10 and Codex-L, but remains clearly visible on FB15k-237 and Hetionet.

The benchmark table also makes the central trade-off of RuleDep more visible. On the three standard datasets shared with prior work, the RuleDep-default configuration and its stronger best or ensemble variants remain competitive with the strongest interpretable baselines, while the Canonical baseline reproduction remains a strong rule-only reference point. This supports the interpretation that canonical aggregation already captures most marginal rule effects. RuleDep adds a smaller but still useful dependency-aware correction on top of that. On the four additional datasets, the new Max+/Noisy-OR rows establish strong application baselines, while the RuleDep-default rows show that the dependency-

aware correction remains beneficial beyond the classic three-dataset benchmark.

The accuracy gains in Table 2 must be read together with training cost. In several datasets, the canonical global model remains a very strong rule-only baseline, which confirms that most predictive signal is already contained in reweighted individual rules. However, the global training regime is substantially more expensive. Figure 1 visualizes this trade-off for six datasets. Hetionet is excluded because the canonical run terminated with out-of-memory errors before producing a comparable wall-clock result.

A similar pattern emerges when we compare against the learned Dense and Sparse aggregators reported by Ott et al. [?]: their canonical logistic baseline is already competitive in accuracy, but the latent aggregators are much more expensive. On FB15k-237 and WN18RR, their logistic model trains $27\times/39\times$ faster than Dense and $4.8\times/7.3\times$ faster than Sparse. Our results extend this efficiency argument from the global rule-only setting to the dependency-aware setting: RuleDep sacrifices some of the accuracy of a single global model on several datasets, but in return it keeps training tractable on datasets where a global dependency-aware model would be prohibitively expensive.

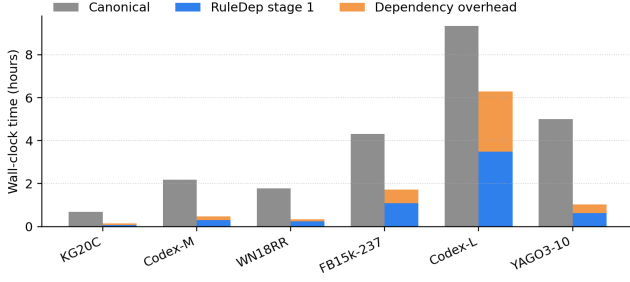


Figure 1: Wall-clock comparison between the global canonical baseline and the best RuleDep configuration on six datasets. The gray bar is the total runtime of the canonical global model. The RuleDep bar is decomposed into the relation-wise canonical baseline sweep (blue) and the additional dependency training overhead (orange). Hetionet is excluded because the canonical run terminated with out-of-memory errors.

Table 3: Representative relation-level gains from RuleDep. Each dataset shows the two relations with the highest relative MRR gain over the canonical baseline; DrD and DpS denote “Disease resembles Disease” and “Disease presents Symptom”.

Dataset	Relation	Avg Obj.	Avg Subj.	Canon.	RuleDep
KG20C	author_write_paper	1.54	2.52	.231	.248
	paper_in_domain	3.61	9.24	.116	.118
WN18RR	_has_part	2.43	1.21	.205	.219
	_member_meronym	2.39	1.01	.318	.329
Codex-M	P20 (place of death)	1.00	13.17	.327	.336
	P737 (influenced by)	2.82	2.47	.119	.121
FB15k-237	combatants	9.11	8.59	.255	.325
	region	1.17	7.48	.573	.657
Codex-L	P749 (parent org.)	1.07	2.64	.275	.293
	P737 (influenced by)	2.58	2.29	.114	.119
YAGO3-10	dealsWith	5.92	7.75	.265	.325
	diedIn	1.00	4.89	.180	.201
Hetionet	DrD	4.11	4.27	.263	.309
	DpS	20.35	6.52	.233	.241

1.3 When Does Dependency Help?

The average gains in Table 2 hide a much stronger relation-level pattern. Across all 430 evaluated relations, 95 relations obtain more than 3% relative MRR gain over the canonical baseline, 218 relations fall into a stable 0–3% gain range, and 131 relations show negative transfer. RuleDep is therefore not a uniformly strong bias, but a selective correction that is useful when the structural conditions are right.

The strongest relation-level gains are substantial: they reach 27.5% on FB15k-237, 22.8% on YAGO3-10, and 17.7% on Hetionet. At the same time, the positive cases are not confined to one dataset family. KG20C and WN18RR show modest but consistent gains on compact structural relations, while Codex-M and Codex-L still contain clear positive cases even when only a few high-precision dependencies survive filtering.

Two trends are especially consistent. First, positive-gain relations are typically not yet saturated under

Table 4: Ablation results across seven datasets. The default row reports MRR; all other entries report $2 \times \Delta\text{MRR}$ in percentage points relative to the default, so negative values indicate degradation. The Avg column averages each row under the same scale. Dash entries indicate unavailable runs.

	Approach	KG 20C	WN 18RR	Codex -M	FB15k -237	Codex -L	YAGO 3-10	Hetio -net
Group A	default	.233	.501	.342	.348	.334	.575	.362
	w/o synergy	-0.62	-0.21	-0.07	-0.20	-0.18	-0.14	+0.02
	w/o redundancy	+0.02	-0.19	-0.20	-0.01	-0.04	+0.02	+0.10
Group B	fix-lift	-0.54	-0.31	-3.38	-4.96	-1.98	-5.23	-2.00
	random-init	+0.18	+0.07	-0.82	-3.52	-0.18	-1.25	-1.01
	surprisal-func	-0.16	-0.07	-0.12	+0.03	-0.14	+0.01	+0.61
Group C	dep-scale	-0.04	-0.13	+0.10	+0.25	+0.05	+0.52	-0.36
	sign constraint	+0.14	-0.20	-0.12	-2.83	-0.14	-0.57	-0.01
	auto-ratio	+0.13	-0.20	-0.48	+0.54	-0.70	-0.63	+0.63
Group D	RD	-1.02	+0.33	+0.01	-0.81	-0.13	-0.32	+0.45
	R2D3	-1.14	-0.26	+0.09	+0.67	-0.37	+0.26	+1.55
	R3D6	-0.90	-0.27	+0.16	+0.64	-0.36	-1.33	+1.34
Group E	unconstrained	-0.02	-0.06	-0.01	-0.02	0.00	+0.00	+0.00
	1x rule	-0.01	+0.02	+0.07	-0.28	+0.00	+0.06	0.00
	4x rule	-0.02	-0.06	-0.01	-0.08	0.00	+0.00	0.00

the canonical baseline: their average canonical baseline MRR is 0.296, lower than the 0.406 observed for negative-gain relations. Second, negative-gain relations actually have higher dependency density on average (6.53 dependencies per rule vs. 1.27 for positive-gain relations), which suggests that raw dependency count alone does not predict benefit—it is the quality and structural alignment of dependencies, rather than their sheer number, that determines whether the correction helps.

The representative relations in Table 3 also reveal a clear semantic pattern. FB15k-237 and YAGO3-10 are dominated by event-participation, role-filling, and contextual relations such as *combatants*, *region*, *dealsWith*, and *diedIn*; in these cases, several partially informative rules must co-occur before a candidate becomes convincing. Hetionet shows the same phenomenon in biomedical form: DrD and DpS benefit because disease-level predictions are naturally supported by multiple corroborating biological paths. By contrast, the Codex datasets contain smaller but still consistent gains on attribute-like relations such as P20, P737, and P749, where only a few dependencies survive filtering but still improve ranking. KG20C and WN18RR lie between these extremes: their positive cases are fewer, but the gains are still concentrated on relations where multiple structural cues agree on the same answer. Overall, dependency helps most when relation semantics make co-support meaningful rather than accidental.

1.4 Ablation and Variant Comparison

We next analyze which parts of RuleDep are actually responsible for the gains. We evaluate fifteen settings grouped into five categories: **(A)** Synergy and redundancy ablation, testing whether each dependency po-

larity is individually necessary (corresponding to §??); **(B)** Dependency-aware aggregation mechanism, validating the core design of learned weights, lift-based initialization, and link function choice (corresponding to §??); **(C)** Training parameters, examining regularization strength, sign constraints, and auto-ratio selection; **(D)** Typed structural sharing, assessing whether grouping rules and dependencies into coarse semantic families improves aggregation (corresponding to §??); **(E)** Dependency quantity, studying the effect of filtering thresholds and per-rule budget limits (corresponding to §??).

Group A: Synergy vs. Redundancy. We isolate dependency polarity by comparing the default model against variants without synergy and without redundancy. Removing synergy lowers MRR on six of seven datasets, and the scaled differences make these otherwise small drops visible. Removing redundancy also lowers MRR on most datasets, but the effect is even smaller and is slightly reversed on KG20C, YAGO3-10, and Hetionet. This suggests that the dependency correction is sparse and conservative: either polarity alone can often recover most of the small global gain, while the two signals remain more informative in relation-level cases than in aggregate MRR.

Group B: Aggregation Mechanism. We validate the three core design choices of the dependency-aware aggregation model from §??. First, the fix-lift variant uses the lift-initialized dependency scores before stage-2 training and consistently trails the learned variants, establishing that lift alone is too noisy and uncalibrated for discriminative ranking. Second, the random-init variant is substantially weaker on the larger datasets, showing that learned dependency calibration benefits from a useful structural prior. Third, the surprisal-func variant remains close to the default, confirming that ranking quality depends primarily on the surprisal-space separation rather than the specific link function. We therefore adopt the sigmoid for its numeric stability without sacrificing accuracy.

Group C: Training Parameters. We study how dependency weights are regularized and constrained during optimization. The dep-scale variant applies additional scaling to the dependency weights and remains close to the default, indicating that the default regularization strength is already well calibrated. The sign constraint variant forces all dependency weights to remain non-negative; it is unavailable on YAGO3-10 and drops noticeably on FB15k-237 (0.334 vs. 0.348), showing that allowing negative weights is important for redundancy suppression. The auto-ratio variant automatically tunes the dependency-to-rule ratio per relation and performs close to default, suggesting that the fixed $2\times$ budget is a reasonable default but not strictly optimal for every dataset.

Group D: Typed Structural Sharing. These rows test whether grouping rules and dependencies into coarse semantic families (as described in §??) improves aggregation beyond the untyped default. The effect is strongly dataset dependent: RD leads on WN18RR (0.502) and is competitive on Codex-L, while R2D3 is best on four datasets (FB15k-237, YAGO3-10, Hetionet, and Codex-M). The finest granularity (R3D6) is rarely best, suggesting that overly specific type splits dilute the shared signal. This confirms that type-level calibration is useful, but only when the grouping granularity matches the dataset’s relation structure. Beyond accuracy, typed models also improve interpretability: different datasets clearly prefer different rule families—Uc-type rules dominate KG20C and the Codex datasets, while U-type rules dominate WN18RR, FB15k-237, YAGO3-10, and Hetionet. The same pattern holds for dependency families, with BB important on WN18RR and Codex-L and UU dominating FB15k-237, YAGO3-10, and Hetionet. Relation-level analysis shows substantial within-dataset heterogeneity, meaning that type-level parameters express relation-specific structural preferences rather than a single global prior.

Group E: Dependency Quantity. We study filtering and budgeted selection strategies jointly. Our default pipeline enforces a $2\times$ dependency-per-rule budget, so we report deviations from that default. Removing the budget entirely (unconstrained) matches the default on several datasets but brings negligible overall gains, indicating that valid dependencies are sparse and ranking performance saturates early. Even when limiting the budget further to $1\times$ rule or expanding it to $4\times$ rule, the effects are extremely minor. This confirms that the strict budget acts as a principled scale limiter that controls cost without bottlenecking performance, and that smaller lifts in the long tail do not meaningfully alter the final outcome.

Overall, the ablation results reinforce two design principles. First, dependency-aware aggregation should be treated as a correction to a strong rule-only baseline, not as a replacement for it. Second, there is no single best dependency configuration: dataset-specific bias matters, which motivates typed structural modeling and relation-wise model selection.

1.5 Learned Weights and Case Study

Finally, we inspect what the model actually does with the dependency parameters after training. The short answer is that RuleDep behaves like a sparse correction model. In every dataset, dependency weights are generally sparser than rule weights after model selection. The near-zero ratio of rule weights ranges from 10.7% to 43.8%, while that of final dependency weights ranges from 62.0% to 98.9%. On most datasets the dependency near-zero ratio exceeds the rule near-zero ratio, confirming that only a small fraction of dependency edges carries non-trivial weight. At the same time, the maximum absolute dependency weight can still be substan-

Table 5: Learned weight behavior by dataset. “R” and “D” denote rule and dependency components; “-nz” and “-max” denote near-zero ratio and maximum absolute value; “#v” and “Top3” summarize valid terms and top-3 concentration.

Data	R-nz	D-nz	R-max	D-max	#vR	Top3R	#vD	Top3D
KG20C	15.8%	62.0%	3.06	0.03	4.28	87.0%	0.00	100.0%
WN18RR	10.7%	83.9%	4.25	0.47	13.34	75.4%	0.13	92.3%
Codex-M	11.4%	96.4%	2.99	0.84	26.69	61.7%	0.81	87.2%
FB15k-237	15.9%	75.6%	5.07	0.56	72.03	31.3%	2.26	68.6%
Codex-L	32.0%	83.1%	4.33	2.94	13.47	68.7%	2.46	86.8%
YAGO3-10	43.8%	98.9%	5.43	1.53	8.82	81.3%	1.42	99.0%
Hetionet	34.1%	97.2%	5.63	0.33	57.74	29.2%	0.69	86.8%
Average	23.4%	85.3%	4.39	0.96	28.05	62.1%	1.11	86.8%

tial, reaching 6.77 on Codex-L and 3.25 on YAGO3-10. This is exactly the intended behavior: most dependency edges should be ignored, while a small number of strong edges act as decisive corrections.

The polarity analysis tells a similar story. Before final model selection, synergy dependencies are more likely to keep positive weights than redundancy dependencies. After selection, both become much more aligned with their intended roles, which again supports the design choice of using mined lift as a structural prior while still allowing supervision to refine or override it.

A representative positive example is the combatants relation in FB15k-237. Here the canonical baseline reaches an MRR of 0.255, and RuleDep raises this to 0.325, a relative improvement of 27.5%. This relation has 28,658 candidate rules and 80,136 filtered dependencies, i.e., a dense interaction space where isolated rule weights are clearly insufficient. A second example is the biomedical relation DrD in Hetionet: its MRR rises from 0.263 to 0.309 with only 583 rules and 182 dependencies. Together, these two cases illustrate the two main modes in which RuleDep helps: either by exploiting large dense dependency pools, or by identifying a small number of high-quality corrections in semantically focused relations.

Table 6 illustrates how dependency correction changes the final ranking for the query `countries_spoken_in(Iraq)` on FB15k-237. Before the dependency layer, English ranks first (score 0.302) and Persian second (0.290). After dependency correction, English is penalized by $\Delta_D = -0.90$ across 484 active dependency terms—202 of which are redundancy—dropping its score to 0.258 (rank 2). The strongest redundancy pair (D484) links R1663645 (location contains, $w = 0.48$) with R59293 (continent/country, $w = 0.33$): both are U-type generalization rules that provide overlapping geographic evidence, so their co-activation is down-weighted.

Persian, by contrast, receives a net boost of $\Delta_D = +0.41$ from 215 purely synergistic dependencies (zero redundancy). The shown B-type dependency pairs illustrate this: D1335 links a propagation-via-adjointing rule (R143831, $s = 0.21$) with a continent-containment chain (R1556835), while D6171 connects an analogous propa-

gation rule (R1036876, $s = 0.21$) with a same-continent rule (R1366646). These closed-path patterns provide specific, non-redundant evidence that Iraq speaks Persian. Because no redundancy applies among these complementary rules, the dependency layer preserves and amplifies their combined contribution through synergy terms.

This single case captures the two regimes in which RuleDep operates: (1) *redundancy suppression* penalizes candidates supported by many semantically overlapping generalization rules, and (2) *synergy preservation* rewards candidates supported by complementary specific rules. The net effect is a rank flip that aligns better with ground truth—Persian is the correct answer for `countries_spoken_in(Iraq)` in FB15k-237.

Overall, the weight analysis and the case studies support the same conclusion as the ablations: RuleDep does not help because it spreads a small bias over all rule pairs. It helps because it learns a sparse set of structurally meaningful corrections on top of an already strong rule-only aggregator.

Table 6: FB15k-237 case study for countries_spoken_in(Iraq). Abbreviations: csi = countries_spoken_in, ol = official_language, lc = location_contains, cc = continents_countries_within, adj = adjoins, cnt = contains, el = event_loc.

Candidate	Dep.	Weight	Left rule	Right rule	Score	Rank
English	D_1	[0.006]	ϕ_1 csi(English,Y) $\leftarrow$ cc(North_America,Y) [0.36]	ϕ_2 csi(English,Y) $\leftarrow$ lc(North_America,Y) [0.53]	.302	1
	D_2	[0.001]	ϕ_3 csi(English,Y) $\leftarrow$ csi(Pakistan,Y) [0.16]	ϕ_4 csi(English,Y) $\leftarrow$ csi(A,Y) [0.45]	$\rightarrow$	$\rightarrow$
	...	482 deps (201 red: -0.40, 280 syn: -0.50, total: -0.90)			.212	2
Persian	D_3	[0.012]	ϕ_5 csi(X,Y) $\leftarrow$ csi(X,A), adj(Y,A) [0.21]	ϕ_6 csi(X,Y) $\leftarrow$ csi(X,A), cc(B,A), cnt(B,Y) [0.002]	.290	2
	D_4	[0.010]	ϕ_7 csi(X,Y) $\leftarrow$ csi(X,A), adj(A,Y) [0.21]	ϕ_8 csi(X,Y) $\leftarrow$ csi(X,A), cc(B,A), cc(B,Y) [0.004]	$\rightarrow$	$\rightarrow$
	...	213 deps (0 red, 213 syn: +0.41, total: +0.41)			.331	1